MSE 360: Kinetic Processes in Materials **Points:** ____/100
S2025
Final Exam

Name (print): ____

I have neither given nor received unauthorized aid on this test.
______________________ (signature)

No notes, books, cell phones, or information stored in calculator memories may be used. Suspicion of cheating will result in the filing of a Report of Academic Integrity Violation. All work must be written on these pages and turned in.

Useful Constants and Equations

$1\text{ eV} = 1.602 \times 10^{-19}\text{ J}$

$1\text{ eV/atom} = 23.06\text{ kcal/mol}$

$R = 1.987\text{ cal mol}^{-1}\text{ K}^{-1}$

$k_B = 1.38 \times 10^{-23}\text{ J K}^{-1}$

$R = 8.3142\text{ J mol}^{-1}\text{ K}^{-1}$

$k_B = 8.616 \times 10^{-5}\text{ eV K}^{-1}$

$R = 82.05\text{ cm}^3\text{ atm mol}^{-1}\text{ K}^{-1}$

$N_A = 6.022 \times 10^{23}\text{ molecules mol}^{-1}$

$e = 1.602 \times 10^{-19}\text{ C}$

$101{,}325\text{ Pa} = 1\text{ atm}$

$$J_i = -D_i \frac{\partial C_i}{\partial x}$$

$$\frac{\partial C_i}{\partial t} = D_i \frac{\partial^2 C_i}{\partial x^2}$$

$$C_i(x,t) = C_i^0 + \left(C_i^* - C_i^0\right) erfc\left[\frac{x}{2\sqrt{Dt}}\right]$$

$$\left(\frac{dx}{dt}\right)_{tot} = -\frac{M}{\rho}\frac{P_{HCl}^0}{4RT\left(\frac{\delta}{D_{HCl}}+\frac{1}{k'}\right)}$$

$$\frac{dx}{dt} = -\frac{M}{\rho}\frac{k' P_{HCl}^0}{4RT}$$

$$\frac{dx}{dt} = -\frac{M}{\rho}\frac{D_{HCl} P_{HCl}^0}{4RT\delta}$$

$$F(t) = 1 - exp\left[-\left(\frac{\pi}{3}\right)\dot{G}^3 \dot{N} t^4\right]$$

$$C_i = C_{i,0}\, exp\left[\frac{\gamma V_m}{RT}\left(\frac{1}{r_1}+\frac{1}{r_2}\right)\right]$$

$$T_m(r) = T_m(\infty)\left[1 - \frac{2\gamma V_m}{r\Delta H_m}\right]$$